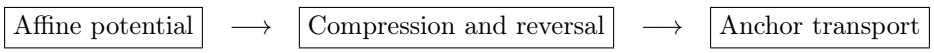
Independent signed graph algebra review

Reviewer /root/monic_review. 2026-11-01.

Result. Nine main declarations, nine producer controls and six independent controls passed fresh isolated checks. All 24 named readbacks use only propext and Quot.sound. No mathematical or union-sign defect found in the scoped algebra claims.

Source-first derivation. Before reading proof bodies, the reviewer derived the native potential invariant $x = sR + a + bk$. Compression is affine composition. After endpoint reduction, an edge is $c_u R_u + c_v R_v = A + Bk$. Involutive signs give $R_u = -c_u c_v R_v + c_u A + c_u Bk$. Reversing the parent/child orientation preserves sign s and transforms offsets by $-s$, matching the native size-swapped branch.



Fixed anchors. A fixed child $x + yk$ and potential $sR + a + bk$ yield parent $s(x - a) + s(y - b)k$. This is the proved transport identity. A second fixed parent anchor supplies a difference equation; a first anchor alone does not determine the target. The packet does not prove native metadata mutation or same-root cycle/inversion branches.

Independent controls. All four sign combinations and both union orientations, positive fixed-child propagation, wrong offset sign, invalid involutive sign, consistent anchor difference and equal-anchor zero-coefficient cases passed. No fresh failures occurred.

Native obligations. Parent forest acyclicity and recursion termination, index safety, root potential identity, path compression and union mutation preservation remain separate. Native signs must stay exactly ± 1; signed() treats every other value as negation. Canonical words and addition headroom $2r - 2 < 2^{64}$ are not supplied by constructor primality alone. Fixed-root metadata, counters, inverse branches and compiled equivalence are also separate. Model fidelity is faithful at scope; whole native graph correctness is not established.

Workflow. Separate reviewer, prospective local plan, source-first derivation, fresh replay and named axiom audit completed. Plan was not precommitted; dispatch inputs were stable but not published immutably. Imported compiled helpers remain a trust boundary; full dependency rebuild, independent kernel checker, independence gate and publication are coordinator obligations. No scientific runs or official transition occurred.

Detailed evidence: REVIEW.md, status.json, attestation, receipts, bindings and complete logs. Native reference: three_leads.rs lines 1491–1530 and 1558–1712. Editable flow.svg accompanies this report.